

**Table 3-3: Lmax and Cross-section Area When CT Secondary Current is 5A and under Single Module System**

| Cross-section Area of the Conductor | Lmax    |         |         |         |         |         |
|-------------------------------------|---------|---------|---------|---------|---------|---------|
|                                     | 15VA CT | 20VA CT | 25VA CT | 30VA CT | 35VA CT | 40VA CT |
| 2.5 mm <sup>2</sup>                 | 36 m    | 50 m    | 64 m    | 78 m    | 92 m    | 106 m   |
| 4 mm <sup>2</sup>                   | 58 m    | 80 m    | 103 m   | 125 m   | 148 m   | 157 m   |

### 3.6.2 Basic CT Installation & Wiring

- The CT for current detection shall be located on either grid side or load side.
- When it's an open loop, the CT for current detection shall be located on the load side to feed the detection signal to the APF as shown in Figure 3-27. When it's a close loop, the CT for current detection shall be located on the grid side to feed the detection signal to the APF as shown in Figure 3-28.
- A set of three CTs must be provided for current detection of the harmonic source.
- The CTs shall be oriented accurately. The default is that P1 is placed toward the grid, and the direction of all CTs must be the same.
- The phase sequences of the detection signal of the CTs must not be changed.
  1. The secondary output S1 of CT for R-phase detection must be connected to the terminal board R\_S1, and the S2 outgoing line must be connected to the terminal board R\_S2.
  2. The secondary output S1 of CT for S-phase detection must be connected to the terminal board S\_S1, and the S2 outgoing line must be connected to the terminal board S\_S2.
  3. The secondary output S1 of CT for T-phase detection must be connected to the terminal board T\_S1, and the S2 outgoing line must be connected to the terminal board T\_S2.



**NOTE:**

1. When compensating a three-phase system, single CT detection can be set up (only the R-phase CT is connected).
2. Corresponding setting shall be set up in accordance with CT wiring under compensation mode; otherwise, the compensation will not be working.
3. The CT wiring shall be fastened to avoid open circuit, which will lead to high-voltage risk and secondary circuit burning.